

Assignment I

AI Compute Platforms, Applications and Trends

Submission Include:

Student Name:

Student ID:

Class:

Assignment Title:

GitHub Repository Link:

Question I.

Create a simple Student Grade Program using Python or Java and manage the project using Git and GitHub.

Instructions :

1. Create a folder named: student-grade
2. Open Git Bash and navigate to the student-grade folder.
3. Initialize the folder as a Git repository.
4. Create a Python or Java program that:
accepts a student's name;
accepts marks for two subjects;
calculates the average mark; and
displays the student's name and average mark.
5. Check the status of the Git repository. Go to GitHub and create a new repository named: student-grade
6. Verify the remote repository connection.
7. Push the local repository to GitHub.
8. **Remote Repository Modification.** Open the program file directly from GitHub. Modify the program on GitHub so that it also displays whether the student has Passed or Failed.
9. Commit the modification directly on GitHub.

10. Local Repository Modification Modify the local program so that it also calculates and displays the Total Mark of the two subjects. Example: Total Mark: 150
11. Add the modified file to the staging area.
12. Commit the modification.
13. Try to push the local changes to GitHub.
14. Observe the result and explain why the push cannot be completed immediately.
15. Retrieve the latest changes from GitHub.
16. Integrate the remote changes with the local changes. Resolve any merge conflict if one occurs. Make sure the final program displays:

Example final output:

Enter student name: Aung Aung

Enter first subject mark: 70

Enter second subject mark: 80

Student Name: Aung Aung

Total Mark: 150

Average Mark: 75

Result: Pass

17. Commit the final version if necessary.
18. Push the final version to GitHub.
19. Verify that the local repository and GitHub repository contain the same final version of the program.

LOCAL

GITHUB

grade.py

grade.py

Average Mark

Average Mark

_____ Same _____

Screenshots:

Screenshot 1 – Local Repository

Show:

- the student-grade folder;
- the source-code file; and
- the Git Bash terminal.

Screenshot 2 – Git Initialization and First Commit

Show the Git commands used to:

- initialize the repository;
- check repository status;
- add the file; and
- commit the file.

Screenshot 3 – Commit History

Show the output of:

`git log --oneline`

Screenshot 4 – Remote Repository Connection

Show the output of:

`git remote -v`

Screenshot 5 – First GitHub Push

Show the GitHub repository containing the original Student Grade Program.

Screenshot 6 – GitHub Modification

Show the modification made directly on GitHub for the Pass/Fail result.

Screenshot 7 – Local Modification

Show the local program containing the Total Mark modification.

Screenshot 8 – Push Result

Show the Git Bash output when attempting to push after both the local and remote repositories have been modified.

Screenshot 9 – Pull / Merge Process

Show the commands and results used to retrieve and integrate the remote changes.

Screenshot 10 – Final Program

Show the final program running successfully with:

Student Name

Total Mark

Average Mark

Pass / Fail Result

Screenshot 11 – Final GitHub Repository

Show the final source code in the GitHub repository.

Screenshot 12 – Final Commit History